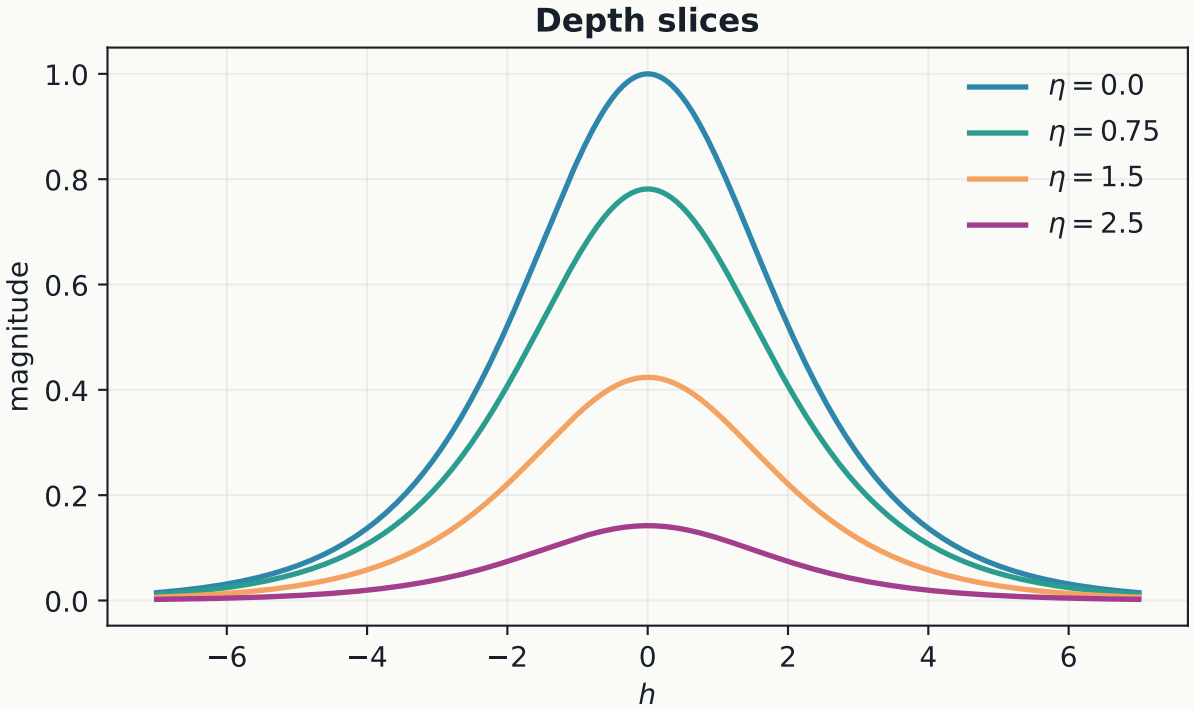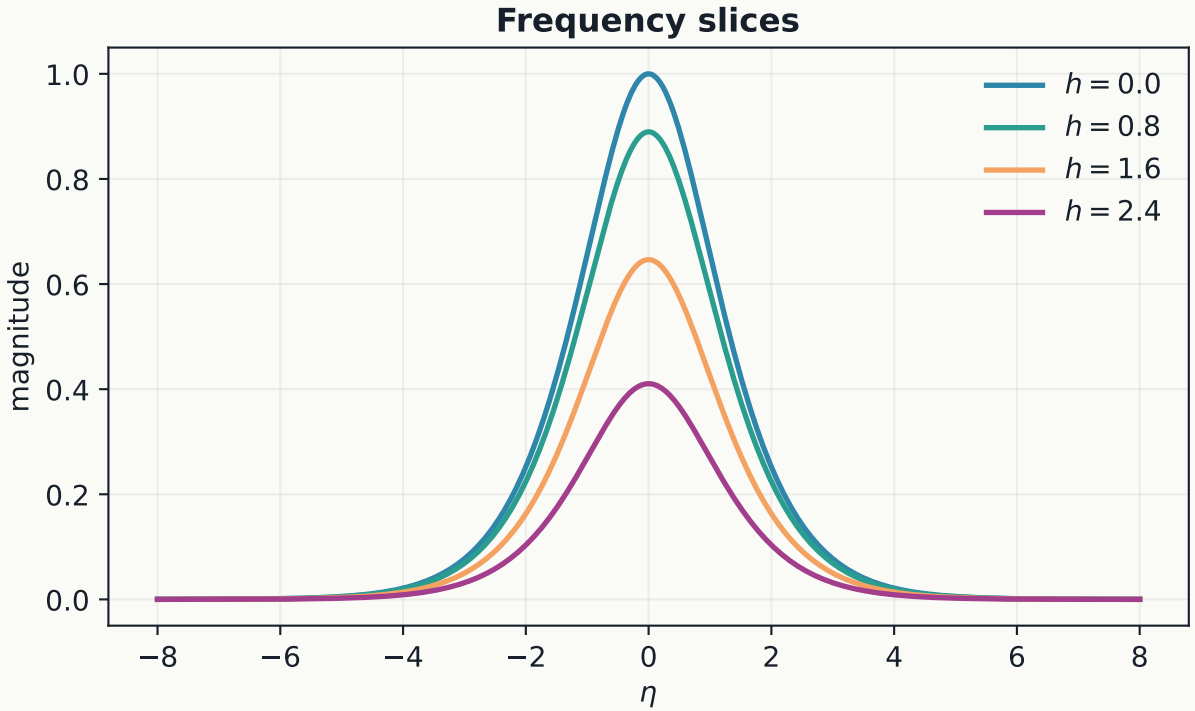The ambiguity magnitude factorizes into depth and frequency envelopes